Aarav Sunil Jadav

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Summary

Tech Lead and full-stack engineer driving backend and data infrastructure for a multi-tenant security platform serving **30+ enterprise customers**. Lead a team of **7+ engineers**, rebuilt log ingestion in Rust to sustain **100k logs/sec** with **33% lower compute cost**, and migrated storage to ClickHouse, cutting storage cost **68%** while handling **10 TB/day** ingest. Own end-to-end distributed systems architecture, cloud deployment, observability, and 24/7 production support. Seeking to build secure, high-scale data systems.

Experience

Eventus Security – *Software Engineer (Tech Lead), Foundation Core* India | Jun. 2024 – Present

- Tech lead for **7+ engineers** on a Kubernetes-based, multi-tenant SOC platform serving **30+ enterprise customers**; own architecture, releases, and **24/7 on-call** support.
- Rebuilt log ingestion from **JVM services in Rust**, replacing network hand-offs with a memory-mapped circular buffer and disk spill under backpressure; sustained **100,000 logs/sec** without GC pauses or data loss, using **33%** less compute.
- Led the migration from **OpenSearch to distributed ClickHouse**, designing ReplicatedMergeTree tables with ZSTD codecs, LowCardinality encoding, typed JSON fields, and bloom-filter skip indexes; cut **storage cost by 68%** and CPU by 10% while supporting **10 TB/day** ingest and sub-second search over billions of events.
- Independently built the platform's **IAM service** end to end: AWS IAM-style allow/deny policies evaluated per request, REST and gRPC APIs, and enterprise federation through SAML, OIDC, and LDAP.
- Developed an **AI-enabled SOAR platform** for alert enrichment, playbook response, and case merging; similarity scoring auto-closed duplicate and low-priority cases, increasing cases handled per analyst by **75%**. Added a RAG-based query interface for threat hunters.
- Built **UEBA in Python** using Isolation Forest, autoencoder, GNN, and Random Forest models for user and entity activity; increased detection coverage by **25%** and reduced false positives by **82%**.
- Migrated CI/CD from Jenkins to **GitHub Actions**; wrote pipelines deploying SOAR and UEBA on **GCP** and IAM and other services on **Azure**, provisioning instances with Terraform and configuring them with Ansible.
- Built a Prometheus-style observability service for health, ingest throughput, queue depth, backpressure, and per-tenant usage, with a **TypeScript/React dashboard** and threshold alerts used by the on-call team.

Open Source Contributor Remote | Jun. 2025 – Present

- Created and maintain SpecMatrix, an open-source Rust compatibility tester for observability backends.
- Contribute fixes across ClickHouse (partition safety), GreptimeDB (PromQL and auth), DuckDB (hash joins), Apache Arrow Rust (decimal statistics), and Apache DataFusion (expression evaluation); continue contributing to other open-source projects.

FluentGrid Limited – *Intern* India | Jan. 2024 – May 2024

- Analyzed **5M+** gas, electric, and water meter records to inform regional resource allocation; built an ML-powered barcode reader that increased collection accuracy by **70%** over the earlier workflow.

Mindarray Systems Pvt Ltd (Motadata) – *Intern* India | Jul. 2023 – Dec. 2023

- Built a **Java query system** executing complex filters over **1M+ records** in **<2 ms**; developed a Python/C# log parser for real-time ML-based categorization.

Education

B.Tech., Computer Science Engineering – Indian Institute of Technology, Goa 2020–2024

Technical Skills

- **Languages:** Rust, Python, Java, TypeScript, JavaScript, C++, C
- **Databases & Query:** ClickHouse, PostgreSQL, OpenSearch, SQL, Qdrant, MongoDB
- **Full-stack:** React, Node.js, REST APIs, gRPC, FastAPI, .NET
- **Infrastructure:** Kubernetes, GCP, Azure, GitHub Actions, Terraform, Ansible, Prometheus, Docker, OCI, Linux, Git
- **AI & ML:** Anomaly detection, RAG pipelines, PyTorch, TensorFlow

Projects

SpecMatrix — Open-source observability compatibility tester (Rust) Sep. 2026 – Present

- Built and maintain a Rust tool that sends edge-case log, metric, and trace payloads to multiple backends, reads them back, and reports silent data changes alongside rejections and exact matches.

Distributed SIEM Log Store and Query Engine (Rust) Jul. 2026 – Present

- Building a SIEM log store with tenant-aware sharding, replicated ingestion, compressed columnar storage, indexed search, automatic promotion of frequently queried JSON fields to typed columns, end-to-end policy-based access control, and a TypeScript/React investigation dashboard.

Automated Correction of Programming Errors Using Deep Learning Feb. 2024 – May 2024

- **GRU/LSTM** models for automated program repair, reaching **95% accuracy** on a purpose-built dataset of 50,000+ C code samples with syntactic errors.